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The temperature profile within Er^{3+}: Yb^{3+}: $LuAl_3(BO_3)_4$ crystal: Insights uncovered by a combined theoretical and experimental study

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ABSTRACT

This work reports a measured temperature profile along the pump light beam inside the Er^{3+}:Yb^{3+}: $LuAl_3(BO_3)_4$ crystal and the simulated ones. The temperature profile featured by a high and asymmetrical peak structure shows rather localized nature with respect to its transversal distribution. By scrutinous examination of the discrepancies between the experimental and simulated ones, a) a heat deposition mechanism elucidating the steady state of local heat generations and various heat transfers was suggested; b) a three-beam of lights model together with a formula for analyzing the fractional thermal load was developed, which leads to a good agreement between the simulated and experimental temperature profiles. The new formula provides a more transparent tool in physics to evaluate the various energy losses; The model can easily be adapted to solve the thermal problem in a laser medium, in particular when a medium show unignorable absorption effect for the laser signal.

1. Introduction

Diode-pumped solid-state lasers in the end-pumped configuration own the advantages of high efficiency, compactness, and reliability [1,2]. In the design of such lasers, a careful consideration must be taken to mitigate the thermal effects such as thermooptic aberrations, lensing and birefringence, which lower the efficiency of laser operation, reduce the output beam quality and change the resonating condition [3–5]. The heat generation can be caused by quantum defects [6,7], stimulated emission [8], energy transfer upconversion [9,10], excited-state absorption [11], cross-relaxation [12], and fluorescence process [7] etc. In its extreme case, if extrinsic defects and/or quantum defects [3] exist, these effects may become serious, which lead to local stress fracture, melting, and other damages to the laser crystal. Thus, the thermal management has long been of critical importance in the design of laser systems. In previous experimental studies, usually only the surface temperatures were measured by using infrared camera [5,13]. While a recent study made an important progress in measuring the temperature distribution within a crystal [14]. This makes it possible to study quantitatively the roles of various factors involved in the heat generation process, if theoretical simulations can be done to scrutinize the various models and assumptions.

In principle, the pump power, the absorption coefficients, the dispersion of the light beam, the presence of defects on the light path and other parameters as well as their temperature dependence, if any, may influence the actual temperature distribution within a crystal. Due to the complexity of the problem, model analysis approach has been tried from the beginning up to the recent time to unravel it [15–17]. In the last few years, with the development of computational technique, a number of numerical simulations have been used to study the temperature distributions within the laser crystals [2,5,18–23]. While these investigations obtained qualitative knowledge of the various factors such as pump power, absorption coefficient, the radius of light beam waist and cooling conditions etc. in governing the temperature distribution [2,5,18,20–23], the quantitative knowledge have remained elusive due to the lack of experimental data.

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In this work, we report a temperature profile along the light path through the Er^{3+}:Yb^{3+}: $LuAl_3(BO_3)_4$ (Er:Yb:LuAB) crystal as measured by using the fluorescence intensity ratio technique [14], and a detailed comparison between the experimental result and the simulated ones. A new model of three-beam of lights was proposed based on the combined experimental and theoretical studies, which revealed the quantitative roles of the various lights and material parameters affecting the temperature distribution. Our work provides clues for improving the design of lasers with respect to the thermal problem. The approach can also be used to evaluate a crystal with regard to its thermal parameters, possible impurities as well as self-absorption of laser signals.

2. Experimental method and simulation models

2.1. Experimental conditions

A temperature profile along the pump laser beam through the Er:Yb:LuAB crystal was measured by using the fluorescence intensity ratio technique [14] with the experimental setup shown in Fig. 1. A continuous wave laser of 976 nm generated by a fiber-coupled diode laser with a 100 μm core diameter was used as a pumping source. The pumping beam was focused by using a telescopic lens system into the laser crystal, and had a beam waist diameter of 160 μm. An uncoated Er:Yb:LuAB crystal (5 × 5× 0.74 mm^3) and a closely contacted 2.4 mm thick sapphire plate were mounted on a copper holder cooled with water of ~ 293 K. A film deposited onto the front surface of the sapphire as an input mirror has a transmissivity of 90 % at 976 nm and a reflectivity of 99.8 % at 1.55 μm. An output mirror with a curvature radius of 100 mm and a transmissivity of 1.7 % at 1.55 μm was used. The cavity length was adjustable around 100 mm. The UC fluorescence signal emitting from the 5× 0.74 mm^2 surface of the crystal was first focused by using a convex lens with a focal length of 27 mm and then filtered to remove the scattering signals from the pump light and the laser. The image and the clean fluorescence signals were measured with a point-by-point scan mode by using a high-resolution optical fiber spectrometer (QE6500, Ocean Optics), while the crystal was in a resonating operation for a 1.55 μm laser.

2.2. The geometrical and numerical models

In this work, the experimentally measured Er:Yb:LuAB crystal has a section area of 5 × 5 mm^2 perpendicular to the beam direction, which is approximately three orders larger than the light spot. As the heat generation occurs mainly on the light path, thus a cylinder model having a rotational symmetry around the beam was created to simulate the combined object of sapphire and crystal (Fig. 2). Shown in Fig. 2 is a model consisting of sapphire and crystal wafers having the same

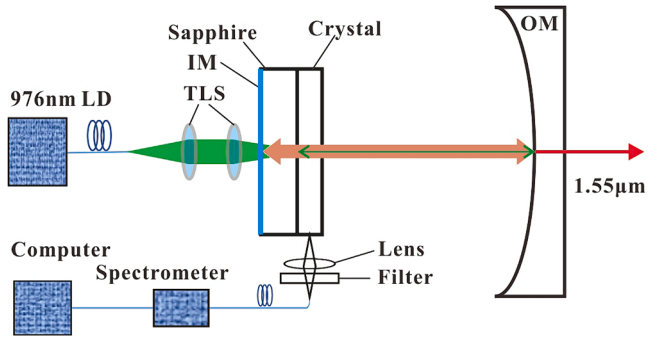


Fig. 1. Sketch of the experimental setup; the thick green line denotes the pump light, and the thin green line indicates the possible residue pump light; the thick and thin red lines indicate the lasers in the resonant cavity and the output one, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

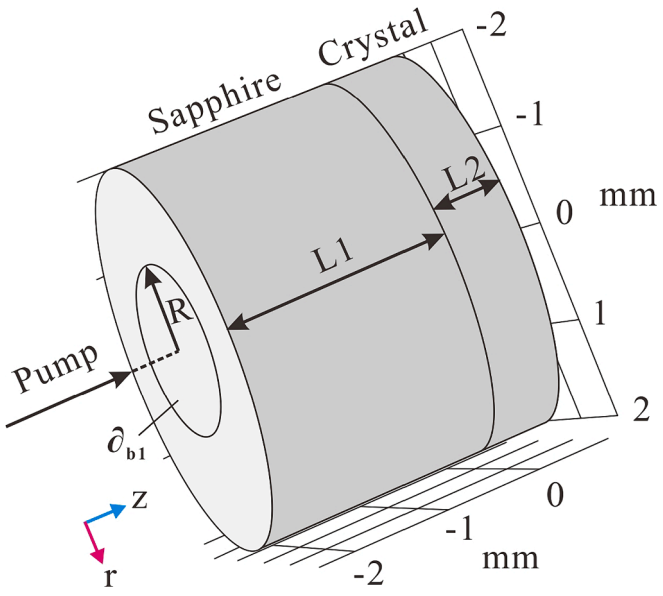


Fig. 2. The geometrical details of the simulation model and the chosen cylindrical coordinate system with its origin locating at the center of the interface between sapphire and the crystal. The surface of the whole sapphire-crystal compound (SCC) is denoted with a symbol ∂, while the two small circular surfaces with R = 1 mm at the two ends of the SCC are denoted by ∂_{b1} and ∂_{b2} for z = -2.4 mm and z = 0.74 mm, respectively.

thicknesses as the measured sample, *i.e.*, L1 = 2.4 and L2 = 0.74 mm, for sapphire and crystal, respectively. The radius of the cylinder model was chosen to be 2 mm. The pump beam enters into the sapphire-crystal model through the ∂_{b1} surface with the output laser leaving the model from the opposite surface ∂_{b2} (see Fig. 2 and Fig. 1).

The heat generation and deposition mechanisms in a gain medium can be rather complicated. In principle, if the photons decay into an appreciable number of phonons (multi-phonon relaxation [24]) through various processes such as fluorescence [7], up-conversion [9,10], excited state absorption [11], cross-relaxation [12], concentration quenching [7] etc., heat results. Such processes may involve the relaxations either from the pump band to the upper metastable level or from the lower laser level to the ground state, or from the dead sites or from other defects, etc. Among these relaxations, the processes involving the laser levels are generally considered as the main source of heating [3,25]. In contrast to the heat generation mechanisms, the deposition mechanism has less been noticed. The traditional thinking emphasizes the role of thermal conductivity of a host material, which does not take into account the microscopic details of the heat transport [26]. This work used a phenomenological steady-state heat equation as follows to probe the temperature distribution in the crystal.

$$\nabla\cdot(k\nabla T) + Q(r,z) = 0 \tag{1}$$

In our simulation, the temperature and space dependence of the thermal conductivity, k, is not considered. And $Q(r,z)$ represents the heat source with a unit of W/m^3, (r,z) is the position coordinate of a point in the cylindrical coordinate system (See Fig. 2).

According to the experimental setup, the whole surface of the sapphire-crystal compound (SCC) model was assumed to have good contacts with heat sinks except for the ∂_{b1} and ∂_{b2} surfaces which are exposed to the air. Since the physical properties of the crystal and the sapphire remain virtually unchanged within the experimental ranges, the important physical parameters used in this work are considered as constants (Table 1).

In this work, as the SCC was mounted on a copper holder with water cooling at ~ 293 K, a perfect thermal contact was assumed for the surface specified by $\partial - \partial_{b1} - \partial_{b2}$ and the heat sink copper (see Fig. 2 and

Table 1
Physical parameters of the Er:Yb:LuAB crystal [27–29] and the sapphire [30].

Material	Density,ρ (kg/m^3)	Specific heat,C_p (J/(kg*K))	Thermal conductivity,k (W/(m*K))
Er:Yb:LuAB Crystal	3840.0	660.0	4.7
Sapphire	3970.0	760.0	42.0

Fig. S18 in Supplementary material (SM)). For this reason, the following boundary condition was assumed for this surface:

$$T = 293K \tag{2}$$

As the surfaces specified by ∂_{b1} and ∂_{b2} were directly exposed to the air, the Robin-type boundary condition as follows was applied to ∂_{b1} and ∂_{b2}, respectively,

$$\mathbf{n}\cdot(-k\nabla T) = h(T_s - T_{air}) \tag{3}$$

where $\boldsymbol{n}$ is the outward normal vector of an interested surface, h is the heat transfer coefficient between the surface, $\partial_{b1}(\partial_{b2})$, and air, which is assumed as a constant in this work, $h = 50W/m^2K$ [31]. T_s denotes the temperature on a surface ∂_{b1} or ∂_{b2}; and $T_{air} = 293K$ is the air temperature.

As discussed above, the heat generation originates from the non-radiative dissipation of the absorbed light energy proportional to the Beer-Lambert factor, $exp(-\alpha z)$, while the heat distribution along the directions perpendicular to the beam path was assumed to follow a Gaussian law, $exp(-2r^2/w_p^2(z))$, due to the local nature of the heat generation. The final form of $Q(r,z)$ as widely used in the literatures [32,33] is given as Eq. (4),

$$Q(r,z) = \frac{2\alpha\xi P_{in}}{\pi w_p^2(z)[1-\exp(-\alpha L)]}\exp(-\alpha z)\exp\left[-2r^2\Big/w_p^2(z)\right] \tag{4}$$

where P_{in}, ξ, α and L denote the power of an incident light, e.g., bump light etc., the fractional thermal loading of a crystal [3], the absorption coefficient of a crystal for a specific light and the thickness of the crystal, respectively (see Fig. 2). The $w_p(z)$ is the beam radius function describing the beam shape of a laser along the beam direction in the crystal [34]:

$$w_p(z) = w_{p0} + \theta_p|z - z_0| \tag{5}$$

Where w_{p0} denotes the waist radius of a beam; θ_p is the far-field half-angle of a beam, z_0 is the focal plane position of a beam. See SM1, SM4, SM7 for more explanations of the simulation parameters).

To understand the experimental temperature distributions, the COMSOL Multiphysics® software was used to solve Eq. (1) with various boundary conditions as described in Eqs. (2)-(3) or otherwise as described in a specific simulation case. In general, if the experimental value for a parameter, e.g., P_{in}, is known, we use it, otherwise we treat it as an adjustable parameter estimated from experimental conditions. The workflow of the numerical simulation and the analysis is given in SM7.

2.3. The evaluation of the simulation results

To evaluate quantitatively the quality of the simulation, we use Eq. (6) which measures pointwise the relative deviation of a simulation curve from an experimental one.

$$R = \sqrt{\frac{\sum w_i[y(z_i) - y'(z_i)]^2}{\sum w_i[y'(z_i)]^2}} \times 100\% \tag{6}$$

Where in Eq. (6), z represents the coordinate along the z axis (Fig. 2), y and y' are the values of the simulated and experimental temperatures at the same coordinate z_i. w_i denotes the weight at point z_i, which can be unit or e.g., a reciprocal of the error at a point, etc. Eq. (6) is widely used as a weighted factor to evaluate the quality of a simulation for an experimental data set in structure science [35].

3. Results and discussion

3.1. Experimental results

Shown in Fig. 3 is one measured temperature profile (Exp-SCC) along the beam path in the crystal with an incident pump power of $P_{in} = 5.43$ W. It is obvious that the lowest temperatures (~427.0 – 412.0 K) occur at the two ends of the beam path, while in the intermediate a broad peak centered at z = ~ 0.24 mm appears. This general feature has also been observed in previous theoretical simulations [2,21,23], however, not yet well understood. In particular, if one takes a close look at the experimental curve, one can find a strong asymmetry of the big blunt peak and a possible dip at its summit together with a low and broad peak occurring at the end of the right tail. In order to optimize the experimental conditions, three independent measurements with different parameters have also been made. To understand the observed experimental features quantitatively, simulations based on different models together with in-depth analysis have been performed. Here, we analyze only the results for the first measurement, the analyses for the rest three are given in SM5.

3.2. Simulation vs. experiment: Main features of the temperature profile

3.2.1. Simulation models and heat transfers at steady state

To explain the observed temperature distribution, we first consider a one-beam laser model as done in many previous studies [2,5,18–23] and different crystal models imposed on various boundary conditions. The used parameters are given in Table 1-2. The results shown in Fig. 4 a) and b) are the transversal and longitudinal temperature distributions obtained from the SCC model with the boundary conditions specified by Eq. (2) and (3) (Sim-SCC). The longitudinal distribution is also shown in Fig. 4 c) with a solid black line (Sim-SCC) together with the experimental result (Exp-SCC). In Fig. 4 c), the result obtained from a bare crystal model without sapphire, but with identical boundary condition as in Sim-SCC imposed on the surface of the model, is also shown (Sim-Cry). Shown in the inset of Fig. 4 c) is the result obtained from the same model as in Sim-Cry, but with a boundary condition specified by Eq. (3) imposed on the whole surface of the model.

As one can see from Fig. 4 a) and b), the temperatures on the light path are obviously higher than those at its surroundings, a phenomenon

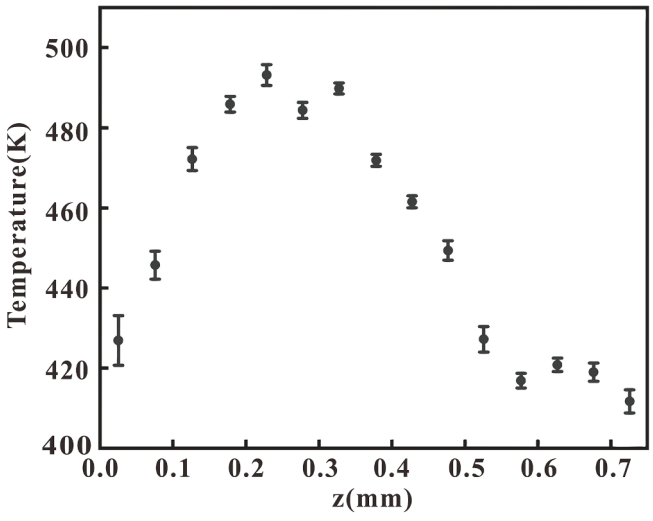


Fig. 3. The measured temperature profile along the pump beam path in Er:Yb:LuAB crystal.

Table 2
The parameters and their values.

Parameters	Value
Incident pump power, P_{in}	5.43 W
Fractional thermal load,ξ	0.40
Waist radius of pump beam,w_{p0}	80.0 μm
Focal plane position of the pump beam, z_0	0.27 mm
Far-field half-angle of the pump beam,θ_p	0.40, ref. [21]
Absorption coefficient of Er:Yb:LuAB crystal, α	4500 m^{-1} (λ = 976 nm), ref. [36]

resulting from the interaction between laser and crystal. This result implies that there is a net deposition of heat when the steady state is reached due to the balance of generated heat and that flowing away from the light path. Intuitively, the much higher temperature on the light path than those of the surrounding copper heat sinks means that the heat flows outward in any transversal directions. However, the situation along the longitudinal direction *z* is more complicated. In view of either the simulated temperature profile (Sim-SCC) or the measured one (Exp-SCC), it is ready to infer that when the system reaches its steady state the heat flows leftward on the left side of the main peak, while on the right side the heat flows rightward, and at the top of the peak the opposite heat flows are all balanced, accordingly the temperature reaches the maximum. Shown in Fig. 4 d) is a scheme illustrating the heat deposition process in the neighborhood of an active center lying at the left side of the main peak. To understand quantitatively the heat transfer process, the heat transferred across the various surfaces denoted in Fig. 2 and Fig. S18 in the SM are calculated, which indicate that a major amount of heat on the light path is transferred from the side surface, while that flowing outside of the surface exposed to the air accounts for only ~ 4.0 % of the total heats flowing out, revealing the importance of the copper heat sinks. The calculated directions of the heat-flows shown in Fig. S19 provided numerical evidence to the heat deposition mechanism shown in Fig. 4 d) (See all details in SM6).

3.2.2. *The overall agreement of main features*

As shown by the compared simulation and experimental results, Sim-SCC vs. Exp-SCC in Fig. 4 c), some main features of the experimental temperature distribution are well reproduced by the simulation. These include mainly: 1) the occurrence of a main peak; 2) the asymmetry of the peak shape; 3) roughly speaking, the position and height of the main peak. More quantitatively, one can also find from the Exp-SCC curve that the temperature nearby the incident end (z = ~ 0.03 mm), T_{in}, is 426.9 K, higher than T_{out} = 411.7 K nearby the output end (z = ~ 0.73 mm). This trend ($T_{in} > T_{out}$) was correctly reproduced by our simulation, which obtained the respective values of 404.6 K and 368.7 K at the corresponding locations, respectively. Some of these features have also been observed in previous simulations [2,21,23], but have never been explained. Extracting the data at 200 uniformly spaced sample points from the experimental and the simulated curves, respectively, we obtained a R value of 6.70 % by using Eq. (6), which indicates an overall agreement of main features between Exp-SCC and Sim-SCC, but not the details.

3.2.3. *The cause for the main peak*

The analysis above shows an overall agreement between Sim-SCC and Exp-SCC. However, the cause for the occurrence of the main peak has remained unexplained.

It is well-known that the pump power decays exponentially in the beam direction of the light due to the absorption (see SM Fig. S1), implying a decreasing trend of the temperature.

However, this is true only in the right side of the peak (see Fig. 4b, c). Our simulation result shown in the inset of Fig. 4c indicates clearly that if the crystal is cooled only by air, the heat deposition becomes more serious, and the peak-like profile does not occur. In fact, as one can see from the simulation curve of Sim-Cry in Fig. 4c, even if the crystal is cooled by cooper heat sinks plus air as described by Eq. (2) and (3), the peak feature is still missing, though the temperatures are overall pulled down dramatically. A simple comparison between the solid line (Sim-

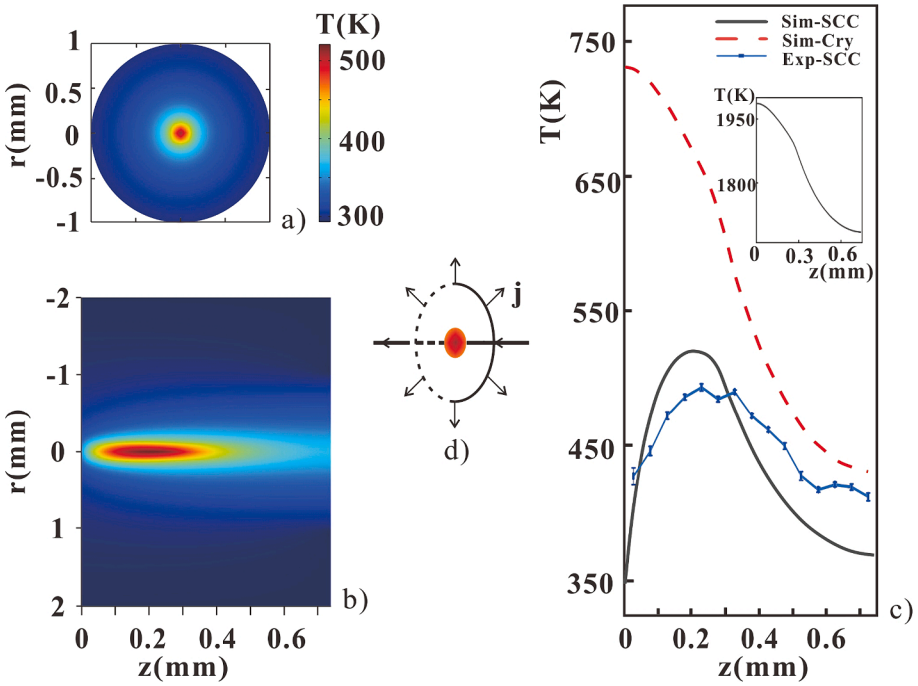


Fig. 4. The simulated results with a one-beam model: a) The simulated temperature distribution on a section perpendicular to the beam direction at z = 0.27 mm (Sim-SCC); b) The simulated temperature distribution in the (r, z) plane with an azimuthal angle $\theta = 0$ in the cylindrical coordinate system (see Fig. 2) (Sim-SCC); c) The simulated results with various cooling configurations compared to the experimental one; d) A scheme illustrating the heat deposition mechanism in the neighborhood of a dopant on the beam z-axis, the red point shows the heat generation, the arrows show the directions of the heat flows. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

SCC) and the broken line (Sim-Cry) in Fig. 4c reveals that it is the presence of the sapphire which results in the peak structure of the temperature profile. To reveal the importance of the sapphire, we simulated a crystal without sapphire by varying the heat transfer coefficient in the Robin-type boundary condition imposed on ∂_{b1} at $z = 0$ (see Fig. S2 (a) and (b) for the difference between Sim-*h*-Cry with Sim-*h*-SCC). The results shown in Fig. S2 (a) indicate that when h is increased the temperature on the center of ∂_{b1} will gradually decrease, and accordingly a peak will be formed at somewhere with $z > 0$ in the crystal. Thus, the ∂_{b1} surface plays a role of a thermal resistor whose effect can be dramatically reduced in the SCC configuration. The generated heat on the beam line can easily be transported to the surface of the sapphire due to its much higher thermal conductivity than that of the air and crystal. As a consequence, the Robin-type condition in the case of SCC becomes insignificant (Fig. S2 (b)). The above analysis has also suggested that the much more rapidly lowering of temperature nearby $z = 0$ on the simulation curve (Fig. 4c) than that of the experimental one may result from the approximation of perfect thermal contact between the sapphire and crystal.

3.3. Simulation vs. experiment: Minor features of the temperature profile

3.3.1. Minor features beyond one-beam model: The role of laser beam parameters

Our discussions in the previous section focus on the main features of the temperature profiles. As one can see from Fig. 4c, the simulated curve for SCC (Sim-SCC) differs from the experimental one (Exp-SCC) in a) the overall lower curve at the right-side of the peak (Sim-SCC) than that of the Exp-SCC plus the missing of a small broad peak in the tail end (Disc-a); b) the height and the position of the main peak (Disc-b). In fact, part of these features has appeared in earlier simulation curves [2,21,23], however, due to either the lack of experimental data or the neglect of these features, the implications of these features have never been explored, which leads to an oversimplified one-beam model and exaggerated fractional thermal load coefficient etc.

According to Eq. (4) and (5), one may expect that the global parameters P_{in}, ξ, α and L may also influence the global features of the temperature profile. However, as P_{in} and L are decided by experimental conditions, they were set as the experimental values whenever possible. The role of ξ and α will be investigated in the later sections. We now discuss the influence of the beam radius function $w_p(z)$. As shown in Eq. (5), this function depends on three parameters, *i.e.*, w_{p0}, the waist radius; θ_p, the far-field half-angle of a light beam, and z_0 the focal plane position. Eq. (5) clearly shows that w_{p0} is a basic parameter for a light beam, and is decided mainly by the telescopic lens system TLS (Fig. 1). Thus, any errors of the TLS and its relative position with respect to the SSC etc. may result in the error of w_{p0}. Fig. 5 (a) shows that the variations of the peak temperature caused by the error, $\Delta w_{p0} = \pm 10\mu m$, is $\sim\mp 12.4K$, which is too small to explain Disc-b, ~27.5 K. As can be seen from Fig. 5 (b) and (c), the changes of both z_0 and θ_p can result in the changes of the peak temperature by $\sim \mp 38.6K$ and $\sim \mp 10.0K$, respectively, within the indicated ranges. We have tried to use a larger range of θ_p, which is a little unrealistic, and obtained a larger temperature change of $\sim \mp$ 21.2 K (Fig. S3), a value still smaller than $\sim$ 27.5 K. It is clear that the decrease of w_{p0}, z_0 and θ_p can all result in an increase of the peak temperature, which is understandable because the decrease of these parameters means thinner beam and higher energy density. However, except for z_0, the resulting changes of the peak temperature due to the other two parameters are not enough to explain the Disc-b. More importantly, the increase of the peak temperature due to the lowering of z_0 is always accompanied by the leftward shift of the main peak thus enlarging the Disc-b instead of reducing it.

The Disc-a is a more challenging problem. As shown in Fig. 5, the largest change of $\Delta T(z) = \sim 2.8$ K at z = 0.73 mm was obtained by changing the parameter z_0 (Fig. 5 b), which is far too small to explain the difference between Sim-SCC and Exp-SCC, $\Delta T =$ 43.0 K at z = 0.73 mm (Fig. 4c), not to mention the low broad peak at the tail of the Exp-SCC curve. This result and those given in the last paragraph indicate clearly that the variations of $w_p(z)$ cannot explain the Disc-a and Disc-b, if the used parameters are confined in the reasonable ranges.

3.3.2. Fractional thermal load evaluated by a new formula and two-beam simulations

To understand these disagreements, one needs to analyze those factors which are not considered in the currently used model. As shown in Fig. 1, the pump light may not be completely absorbed (see also Fig. S1 in SM) in one-time passing through the crystal, the reflected pump light beam exists due to the incomplete absorption and the presence of the OM in the resonating cavity (Fig. 1). Furthermore, the reflected light may not perfectly coincide with the incident pump light beam, which will definitely affect the temperature profile within the crystal. Besides, there may exist other processes of energy loss such as non-stimulated radiation emission etc. The microscopic mechanism of the heat generation in a laser crystal is a rather complicated issue. The previous studies focusing on the various origins have resulted in rather complicated expressions for the fractional thermal load ξ [3,7,9,10,37], which are difficult to evaluate and to apply to calculate the various contributions from different channels. Among these studies, Brown [7] provided a rather comprehensive analysis and suggested a rule to check the reliability of his formulas, that is energy conservation. In this work, we use an approach different from the previous ones. We start from the energy conservation condition and focus on non-thermal channels (See SM1 in SM for details). The result is as follows,

$$\xi = 1 - \eta_c - \frac{N_f}{\eta_P N_1}\cdot\frac{\lambda_P}{\overline{\lambda_f}} - \frac{N_L}{\eta_P N_t}\cdot\frac{\lambda_P}{\lambda_L} \tag{7}$$

Where the calculation of ξ has considered the various energy losses discussed in Sec. SM1 of SM; the percentage of the absorbed energy by

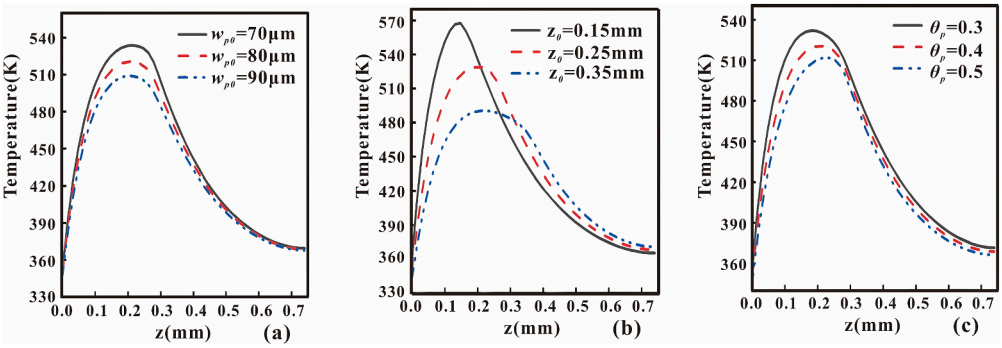


Fig. 5. The influences of the beam radius function $w_p(z)$ due to the variations of (a) the waist radius of the light beam, w_{p0}; (b) the position of the focal plane, z_0; (c) the far-field half-angle, θ_p. The values for unspecified parameters are chosen the same as given in Table 2.

the crystal with respect to the total incident pump energy is described by η_P; the percentage of all energy losses except for the heat and radiation with respect to the absorbed energy is denoted by η_c; N_t is the total number of photons of incident pump laser plus the reflected one with a wave length λ_P; N_f is the number of photons with an average wave length $\overline{\lambda_f}$ in the non-stimulated radiation emission process; N_L is the number of photons with a wave length λ_L produced in the stimulated emission process (laser). The precise mathematical definition of these parameters can be found in Sec. SM1 of SM. It is clear from Eq. (7) that if one assumes $\eta_c = 0$, and neglects the non-stimulated emission, $N_f = 0$, and assume a complete absorption of incident energy, $\eta_P = 1$, the well-known formula [38], $\xi = 1 - \lambda_P/\lambda_L$, can be obtained. Hence, the value estimated from this simple formula represents an upper bound of the thermal effect. The actual value of ξ should be lower than that estimated from $1 - \lambda_P/\lambda_L$. In the current case, it is ~ 0.37, a value widely used in the literatures [21,23]. The value, 0.40, given in Table 1 is only a fitting parameter optimized with respect to the experimental curve. It is larger than 0.37 and thus is unrealistic.

According to the analysis above, one may need to consider the reflected pump light in the simulations. Before we proceed, we want to point out that a multi-beam model was used by Mescia *et al.* [10] to simulate the temperature distribution in a cladding-pumped Er^{3+}/Yb^{3+}-codoped fiber amplifiers. In their case, the forward and backward pump lights are considered together with the radiation from space. Besides, they considered also the z-dependence of the fractional thermal load due to the cooperative upconversion at high Er^{3+}/Yb^{3+} concentration. The z-dependence is not a problem in our case, because the doping concentration of Er^{3+} in our crystal is rather low, ~ 1.5 at%, and in particular our crystal is very thin, 0.74 mm, compared to the ~ 20 m in the fiber amplifiers simulated by Mescia et al.

The difficulty of the one-beam model leads us to consider a second source of heat generation due to the reflected pump light (two-beam model), and to probe the influences of ξ and the beam radius function $w_p(z)$ by varying only the focal plane position z_0. The results are shown in Fig. S5. It is clear that the increase of ξ will lift the overall temperature profile, while the increase of z_0 will shift the main peak rightward and lower temperature profile (see also Fig. 5 (b)). As can be seen from Fig. S5 (b), the problem of Disc-b can be remedied under the condition of $\xi < 0.37$, however, the problem of Disc-a remains, even if $\xi = 0.37$ is used (Fig. S5 (c)). This fact indicates that some important source of heat generation may still be ignored in our simulations.

One concern is that whether the problem of Disc-a can be solved by introducing impurities in the crystal. Our simulations with models containing metal impurities such as Al, or Pt, the constituent elements of the crucible used to grow the Er:Yb: LuAB single crystals, exclude this possibility (See Sec. SM3 for details).

3.3.3. Simulation results based on a three-beam model

On the basis of the above simulations, we propose a three-beam model for describing the heat generation within our crystal, *i.e.*, bm-1: the major component describing the incident pump light; bm-2: the minor component describing the reflected pump light; bm-3: the minor component describing a laser. The parameters given in Table 3 were chosen according to the simulation results discussed above. In particular, the power of bm-2 was calculated based on the absorption of the crystal (see Fig. S1) using Beer-Lambert's law, which resulted in a value of P_{in} = ~ 0.2 W for bm-2. The P_{in} = ~ 89.4 W for bm-3 was estimated from the output power of the device, ~ 0.76 W and the transmittance, ~1.7 % of the OM. The fractional thermal load $\xi = 0.2$ was chosen for both bm-1 and bm-2, while that chosen for bm-3 was ~ 0.009. These two parameters and the others given Table 3 were adjusted but not globally optimized during the simulations. The simulated result with these parameters is shown in Fig. 6. As can be seen from Fig. 6, the problems of both Disc-a and Disc-b are solved almost perfectly, which can also be seen from the calculated very small R value of ~ 0.96 %. The significantly lower ξ values indicate that the actual thermal dissipation of the pump light is much lower than that (~0.4) estimated in previous works [13,18,21,23,31,32], but in good agreement with that (0.22 ~ 0.24) estimated for Nd:YVO_4 by Brown [7]. Though the thermal dissipation of a laser is very low, its heat deposit within the cavity is not negligible. The analyses of the experimental data from the other three measurements are parallel, which gives further justification of our three-beam model (see SM5 for details). One issue is that whether one can simulate the experimental results by ignoring the reflected pump light (bm-2). We have tried to use a two-beam model, bm-1 + bm-3, to simulate the experimental results (Exp-2–4). In all cases, the simulation curves at the output end are too low to account for the measured curves (See Fig. S12-S14 in SM for the details).

Table 3
The values of parameters used for simulating the general features of the temperature distribution.

Parameters	bm-1	bm-2	bm-3
P_{in}	ibid. Table 2	0.2 W	89.4 W
ξ	0.2	0.2	0.009
w_{p0}	ibid. Table 2	40.0 μm	40.0 μm
z_0	0.2 mm	0.65 mm	0.34 mm
θ_p	0.45	0.45	0.30
α	ibid. Table 2	ibid. Table 2	43.4 m^{-1} (1550 nm), ref. [36]

3.4. The temperature and intensity dependence of the absorption coefficient

As the absorption coefficient α is a global parameter (see Eq. (4)), a question may arise that whether its temperature dependence can significantly influence the temperature profile. The temperature dependence of α *i.e.*, $\alpha(T)$, has been thoroughly discussed by Koechner [39] and studied with the Yb:YAG slab crystals [40,41], which revealed an exponentially decayed temperature influence in reference to a deviation from a base temperature. In our case, the temperature variation along the light path is within ~ 80.0 K, so one would not expect a significant influence of T on α. In fact, even if we consider $\alpha(T)$ by using the experimental temperature dependence obtained from Er:Yb:$YAl_3(BO_3)_4$ crystal [42], the simulations with the one-beam and two-beam model still failed to explain Disc-a, which justify our choice of the three-beam model (See SM4 for the results and detailed discussions).

The nonlinear intensity dependence of the absorption on the light intensity, I, suggests also an intensity dependence of the total absorption

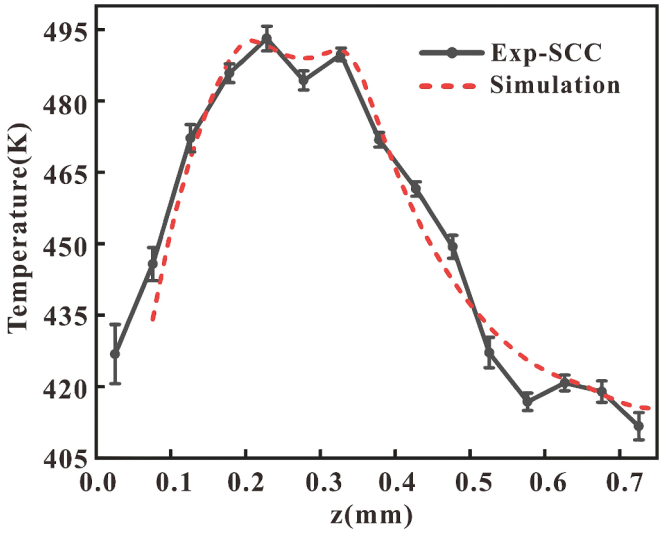


Fig. 6. The experimental temperature profile vs. simulated result based on the three-beam model.

coefficient α_T, *i.e.*, $\alpha_T = \alpha + \beta I + \cdots$ [43,44], where α is the linear part, and β the nonlinear part of the absorption coefficient, which can enter into the Beer-Lambert law as a total absorption coefficient [44]. Such an assignment has also been corroborated by quantum model calculations [45–47]. These studies have also shown that β is a very small quantity at the order of $\sim 10^{-10}$ m/W. Using this value and the values of the power P_{in} given in Table 3 for the pump laser (bm-1) and the laser signal (bm-3) in the three-beam model, one can estimate the nonlinear deviations, βI, as 0.027 and 1.7 m^{-1} for bm-1 and bm-3, respectively, which are much smaller than the corresponding linear coefficients, 4500 and 43.4 m^{-1} (See Table 3). Considering that the intensity of the pump laser decays exponentially with its path length, the actual nonlinear deviation of the absorption coefficient of the pump laser will be much smaller than 0.027 nearby the output end. Hence, the neglect of the nonlinear deviation of the absorption coefficient is a good approximation.

4. Conclusion

Our studies of temperature distribution within the Er:Yb: LuAB crystal by using experimental measurements together with sophisticated theoretical simulations have revealed a very localized character of the heat generation along the light path within the crystal, which is so serious that the crystal can easily be damaged if no good cooling measures were taken. A heat deposition mechanism was proposed to describe the steady state of heat generation and heat transfers. The observed peak structure and large asymmetry of the temperature profile on the two sides of the peak were shown to be the result of the decaying pump energy due to the absorption, and the very different cooling conditions created by the sapphire and air. By scrutinous comparison of the simulated temperature profile with the experimental one, we proposed a three-beam of lights model which takes into account the thermal effects of incident pump light, its reflected residue after absorption and the laser within the resonance cavity. The latter two components were completely ignored in previous works. A formula of fractional thermal load was derived based on the analysis of various possible energy processes within the crystal, with which a reasonable value of the parameter was obtained. Our study indicates that though the absorption of the 1.55 μm laser by the crystal is indeed small, its thermal effect cannot be neglected. The possible effect of impurities was excluded by our simulations with designed impurity containing models. The influence of the temperature dependence of the absorption coefficient was shown to be too small to explain the Disc-a. The simulated temperature profile with the three-beam of lights model and better parameters is in rather good agreement with the experimental one. Our study provides a practical approach to analyzing the thermal effect within an optical crystal, and suggests important factors needed to be considered in designing a laser. The developed approach can also be applied to evaluate a laser medium regarding to its thermal behavior, possible impurities and self-absorption of the laser signals.

CRediT authorship contribution statement

Weiling Fu: Writing – review & editing, Visualization, Validation, Data curation, Formal analysis, Software. **Dawei Qi:** Data curation, Software, Validation, Visualization. **Tingting Zhong:** Writing – original draft, Visualization, Software. **Yujin Chen:** Investigation, Resources. **Yidong Huang:** Supervision, Resources, Investigation. **Shuiquan Deng:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

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